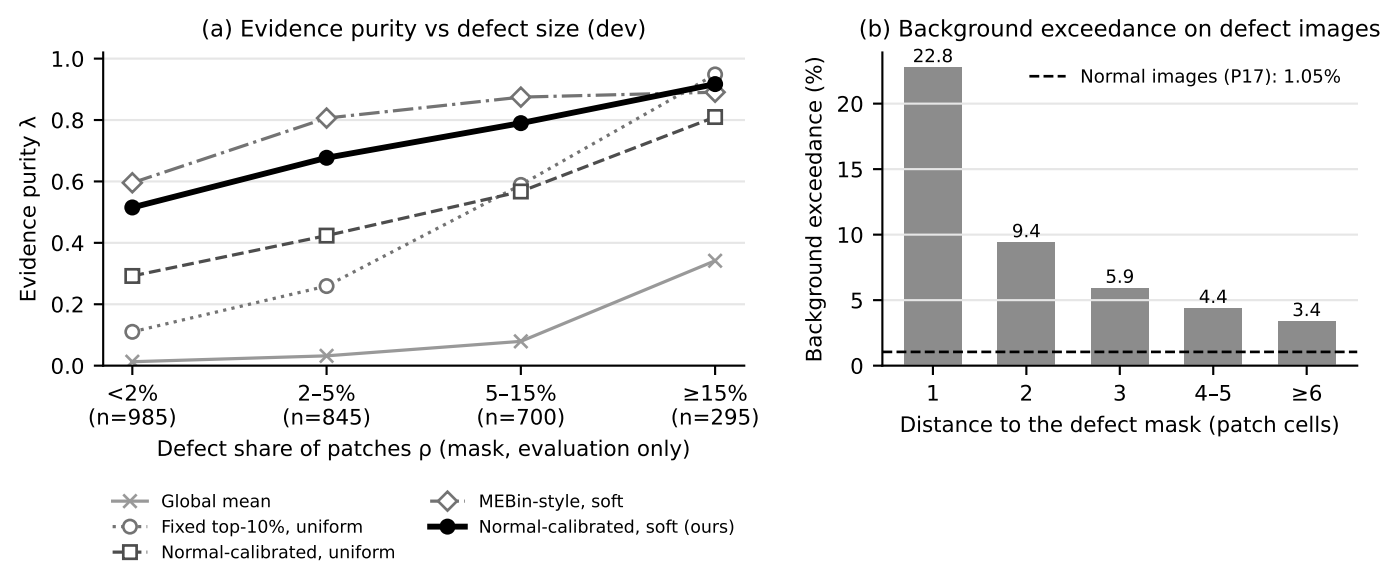
(a) Evidence purity vs defect size (dev)
Evidence purity λ
1.0
0.8
0.6
0.4
0.2
0.0
<2%
(n=985)
2–5%
(n=845)
5–15%
(n=700)
≥15%
(n=295)
Defect share of patches ρ (mask, evaluation only)
Global mean
Fixed top-10%, uniform
Normal-calibrated, uniform
MEBin-style, soft
Normal-calibrated, soft (ours)
(b) Background exceedance on defect images
Background exceedance (%)
Normal images (P17): 1.05%
22.8
9.4
5.9
4.4
3.4
1
2
3
4–5
≥6
Distance to the defect mask (patch cells)